

Blind Angle and Angular Range Detection in Planar and Limited-View Geometries for Photoacoustic Tomography

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Abstract— In this paper, the relationship between the position of ultrasound transducers from the photoacoustic source and the reconstructed image is investigated. Our studies have shown that the distance of the transducer and its location from the photoacoustic source specifically affects and is closely related to the quality of the reconstructed image of the source. Also, we introduce a concept called blind angle for photoacoustic computed tomography (PACT) that have planar or circular limited-view geometry. Complete and accurate equations of this relationship are presented in this paper for all different 2D photoacoustic geometries. The main source of these equations is the spherically of the ultrasound wave produced by the photoacoustic source. The result of these equations is that depending on the location of the transducers with respect to the photoacoustic source, the characteristics of the reconstructed image and the quality of the reconstruction of each point of the photoacoustic source can be accurately obtained, and the reason for these dependencies is fully explained. The results show that the more transducers receive the ultrasound wave produced by the photoacoustic source at more different angles, the higher the quality of the reconstructed image would become.

Keywords— *ultrasound transducers, photoacoustic source, blind angle, photoacoustic computed tomography*

I. INTRODUCTION

Photoacoustic or optoacoustic imaging (OAI) is one of the nascent imaging methods that has received a lot of attention in the last two decades [1,2]. The basis of this imaging system, as its name implies, is a combination of optics and acoustics, in which an optical wave first shines on the tissue under certain conditions, and this radiation causes the desired tissue to heat up, then this warming causes Generating an ultrasound wave from the target tissue [3]. The generated ultrasound wave is received by detectors called transducers and then the original tissue image is reconstructed from the signal received by the transducers. The advantage of this imaging method compared to other methods is that it has high contrast and resolution and it is also a safe imaging method [4,5].

Photoacoustic computed tomography (PACT) [2] is an optoacoustic method which is generally in the form of planar or cylindrical geometries with limited or complete view. In this paper, assuming the geometry is two-dimensional, the arrangement of transducers is considered to be linear and circular. Each geometry has its own applications. Generally, if access from all directions, i.e., 360-degree access to the tissue, is possible, we go for a full-view circular geometry which has many applications in the field of clinical and

preclinical imaging [6-8]. Also, if 360-degree access is not possible, we will inevitably go for a limited-view circular geometry, which is used in optoacoustic tomography of the human thyroid [9] and many other applications [10,11]. Another geometry commonly used in photoacoustic computed tomography (PACT) is the linear geometry, which is widely used in hand-held photoacoustic imaging (PAI) systems for imaging human limbs [12,13] and has many other applications [14,15].

An ultrasound wave is generated by stimulating any part of the tissue with a laser source. According to the Huygens principle, this wave is a circular wave for a two-dimensional geometry and a spherical wave for a three-dimensional environment, and the circumference of this spherical wave becomes larger and larger over the time. Now, if we place several transducers at a certain distance, each of these transducers receives a part of this circular wave, as could be seen in Fig. 1a. Furthermore, given that each transducer is located at a certain distance, l , from the photoacoustic source, it will then receive a signal at a specific time that depends on the values of l , as shown in Fig. 1b. Now, in a time-reversal-based reconstruction, each transducer can be likened to a mirror that emits the received signal as an outgoing circular wave of the transducer at the location shown in Fig. 1c. The reason for propagation in the form of a circular wave in time-reversal reconstruction is because the ultrasound transducer can receive the ultrasound wave at any time in all directions, so only the distance from the transducer is important and cannot differentiate between different directions at the same distances.

Based on the above reasoning, we can conclude that the number of transducers around the tissue should be as high as possible to receive a large part of the ultrasound wave produced by the photoacoustic source and to be able to reconstruct a high-quality image. In the following parts of this article, the relationship between the angle and distance of each transducer from the photoacoustic source and the quality of the reconstructed image of each transducer from a specific photoacoustic source is given in detail. Furthermore, related equations are presented that can be used for structure of each special geometry, to find out which view angles of a photoacoustic source are reconstructed and which parts cannot be reconstructed. These are called blind reconstruction angles of each photoacoustic source. These blind angles are related to the parameters that we refer to in the second part of this paper. Also in the third part, simulations with different

example phantoms are given to validate the presented equations.

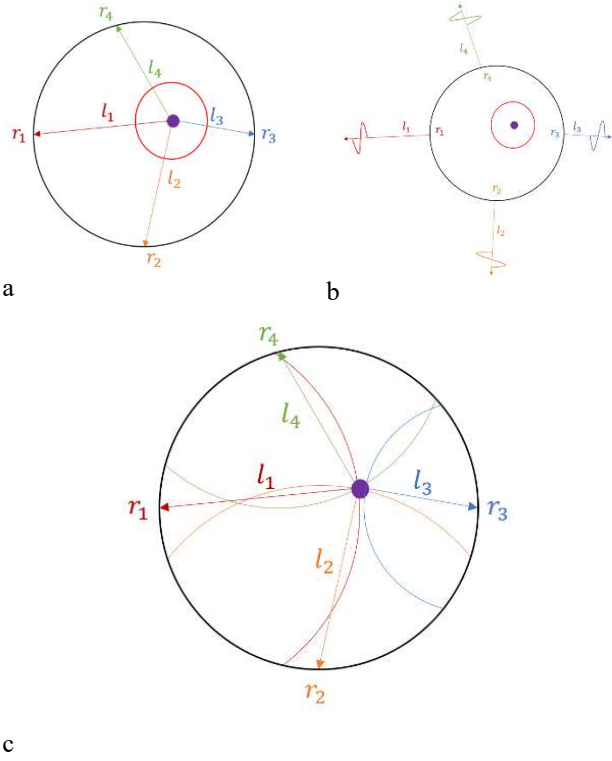


Fig. 1. (a) Four transducers ($r_1 - r_4$) are located around the circle and a photoacoustic source is also generating a circular ultrasound wave. (b) the distance between n th transducer and the photoacoustic source is l_n . So each of these transducers receive the ultrasound wave at specific time. (c) each transducer back-projects the received ultrasound wave into a circular signal.

II. MATERIALS AND METHODS

In this section, we present equations for the dependence of the blind angles for each photoacoustic source inside the chamber on the distance and angle of that photoacoustic source from the transducers. This is done for three different geometries, including planar, limited view circle, and full view circle which are listed in Sections II.A and II.B, respectively. In all simulation, it is assumed that the transducers are point-like and the imaging chamber is two-dimensional.

A. Planar Geometry

In the planar mode, a number of detectors are arranged along a horizontal line as shown in Fig. 2. Fig. 2 shows an arbitrary photoacoustic source generating an outgoing circular wave. Obviously, considering the distance and depth of this photoacoustic source from the transducers, each transducer sees the photoacoustic source from a certain angle and thus receives a certain angle from the generated ultrasound wave, which is plotted in Fig. 2 for the first and last transducer.

The targeted photoacoustic source in Fig. 2 is placed at location (y_s, x_s) . Furthermore, the line connecting the source point and the first transducer makes an angle α with the y axis and θ_1 with the x axis:

$$\theta_1 = \tan^{-1} \left| \frac{y_1 - y_s}{x_1 - x_s} \right| \quad (1)$$

Similarly, the angle of the line connecting the point source and the last transducer makes an angle of θ_2 With the x axis:

$$\theta_2 = \tan^{-1} \frac{D - (y_1 - y_s)}{x_1 - x_s} \quad (2)$$

In the above equation, D is the length of the transducer array. According to Fig. 2, it seems that the location of the transducer at the beginning and end of the array and the distance between the two are very important. The result, in this case, is that the transducers arranged in the planar structure can only receive part of the ultrasound wave produced by the photoacoustic source in an angular range equal to θ as given in (3). These angles start at θ_{start} and end at θ_{stop} , which are given in (4) and (5), respectively.

$$\theta = \theta_1 + \theta_2 \quad (3)$$

$$\theta_{start} = \cot^{-1} \left| \frac{y_1 - y_s}{x_1 - x_s} \right| \quad (4)$$

$$\theta_{stop} = \frac{\pi}{2} - \cot^{-1} \frac{D - |(y_1 - y_s)|}{|x_1 - x_s|} \quad (5)$$

By substituting (1) and (2) in (3) we reach to (6), which indicates that depending on the location of the photoacoustic source relative to the transducers and also the length of the transducer array, it is possible to determine exactly what angular range of ultrasound waves produced by the photoacoustic source is received by the transducers.

$$\theta = \tan^{-1} \left| \frac{y_1 - y_s}{x_1 - x_s} \right| + \tan^{-1} \frac{D - |(y_1 - y_s)|}{|x_1 - x_s|} \quad (6)$$

According to (6), it is concluded that the longer the length D and the shallower the depth of the photoacoustic source, the greater values of θ would be obtained. The relationship between the horizontal distance of the source from the origin is also effective on the value of acceptance angle. The accuracy of the presented analysis for planar mode is validated via simulations in section III.A.

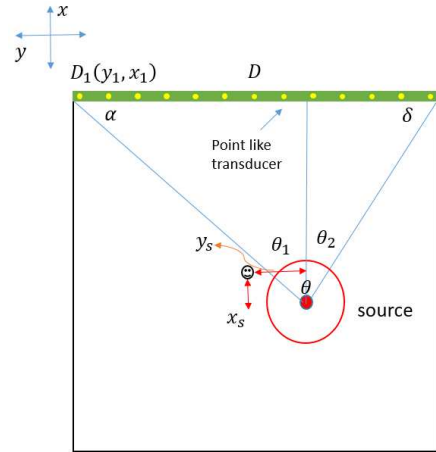


Fig. 2. Outline of a planar geometry. Photoacoustic sources and linear transducers geometry are identified.

B. Circular Geometry

In this section, the transducers are placed on a circle around the photoacoustic source. If we assume that the circle on which the transducers are arranged is not a complete circle and is an arc of a circle which is called a limited-view

structure. In Fig. 3, the schematic of a limited-view structure is plotted with a custom photoacoustic source whose coordinates are indicated by (y_s, x_s) . The center of the circle is assumed to be the origin of the coordinates, and the two transducers at the beginning and end of the arrangement are $D_1(y_1, x_1)$ and $D_2(y_2, x_2)$, respectively, as shown in Fig. 3. In Fig. 3, R is also the radius of the chamber.

The vectors $\vec{S}_1$ and $\vec{S}_2$ in Fig. 3 connect the source to each of the transducers at the beginning and end of the arrangement. These vectors are given in (7) and (8), respectively.

$$\vec{S}_1 = \begin{cases} x_{S_1} = x_1 - x_s \\ y_{S_1} = y_1 - y_s \end{cases} \quad (7)$$

$$\vec{S}_2 = \begin{cases} x_{S_2} = x_2 - x_s \\ y_{S_2} = y_2 - y_s \end{cases} \quad (8)$$

The angle between the two vectors $\vec{S}_1$ and $\vec{S}_2$ is θ_{blind} . Provided that the source is within the range of transducers (left of line D in Fig. 3) and otherwise is equal to $2\pi - \theta_{blind}$. The calculation of θ_{blind} is given in (9).

$$\theta_{blind} = \cos^{-1}\left(\frac{\vec{S}_1 \cdot \vec{S}_2}{|\vec{S}_1||\vec{S}_2|}\right) \quad (9)$$

Operator $\cdot$ Indicates internal multiplication. As could be inferred from Fig. 3, the transducers in the chamber do not receive the ultrasound wave generated by the photoacoustic source at an angle θ_{blind} . We call it blind angle of this photoacoustic source as mentioned in (9). The value of θ , which is the angular range that the transducers receive ultrasound wave from the photoacoustic source located at y_s and x_s , is given in (10).

$$\theta = 2\pi - \theta_{blind} \quad (10)$$

From (10) it can be concluded that the blind angle for each photoacoustic source inside the chamber in limited-view mode depends on the following: limited-view arc size, the coordinates of that point source relative to the origin and the coordinates of the two transducers at the beginning and end of the arrangement, respectively. From the above equations, we conclude that the higher the limited-view angle and the size of the arc of the circle on which the transducers are located, the lower the blind angle range and as a result the higher is the quality of the reconstructed image. The accuracy of the above equations has been verified with simulations in Section III.B.

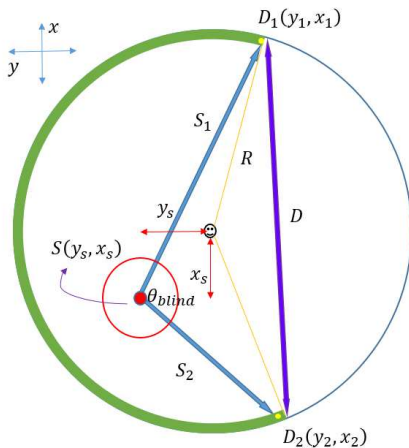


Fig. 3. Outline of a limited-view circular geometry. Photoacoustic sources and limited-view transducers geometry are identified.

If we assume that a completed circular array is considered, it is called full view. With this arrangement, it is expected that all parts of the ultrasound wave generated from the photoacoustic source will be received by the transducers and we will not have any blind angular region. Fig. 4 shows an overview of the full view mode.

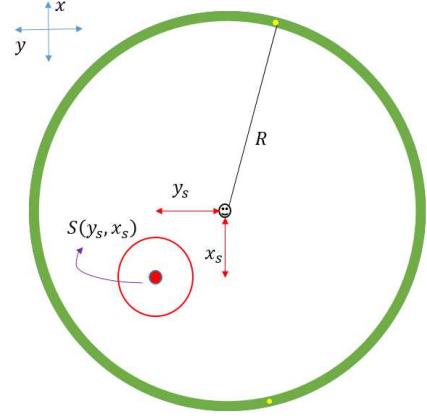


Fig. 4. Outline of a full-view circular geometry.

III. SIMULATION RESULTS

In this section, several simulations for validating the presented analysis are performed. MATLAB K-wave toolbox [16] is used for these simulations. The dimensions of the imaging chamber are different in each part. All detectors are considered point-like with infinite bandwidth.

A. Planar Geometry

In this simulation A two-dimensional chamber measuring 50 by 50 mm has been considered which includes a space measuring 512 by 512 grids. Furthermore, 200 point-like transducers are located on the horizon and above the chamber. The Derenzo Phantom has been used in this simulation. This phantom is shown in Fig. 6a. Fig. 6b shows the reconstructed image. As can be seen, many of the points in Fig. 5a are not reconstructed in Fig. 6b. From this figure, it can be seen that as the depth of the phantom increases relative to the transducers, the quality of the reconstruction decreases. To prove the equations of Part II.A, we derived the acceptance angle from these equations for two arbitrary parts of the Derenzo phantom in Fig. 5b. These parts are enlarged and shown in Fig. 6c and 6d, respectively.

Equation (11) gives the coordinate of the considered photoacoustic source in Fig. 6c. Substituting (10) into (6) will give an acceptance angle of $\theta = 105$ degrees for this source, and also substituting the values of (10) into (4) and (5), will give the values of θ_{start} and θ_{stop} which are 45.4 and 28.6 degrees, respectively. Furthermore, the acceptance angle is calculated as 105 degrees based on (6). In Fig. 5c, if we draw two tangent lines from the center of the photoacoustic source to the edges of the reconstructed angular range of that source, as shown in Figure 6c, the value of the resultant angle would be approximately 105 degrees, which is in good agreement with the derived value from (6). Also, the values of θ_{start} and θ_{stop} are in good correspondence with their derived values from equations (4) and (5).

$$\begin{cases} y_s = -7.5 \text{ mm} \\ x_s = -7.25 \text{ mm} \\ D = 50 \text{ mm} \end{cases} \quad (11)$$

The center coordinates of the photoacoustic points source considered in Figure 6d and the length D are given in (12). By substituting these values in (4), (5), and (6) we will have the acceptance angle $\theta = 89$ and θ_{start} and θ_{stop} of 34.9 and 73.6 degrees, respectively. Approximately the same values could be measured for the high intensity angular region (the angular range received by the transducers) in Fig. 5d.

$$\begin{cases} y_s = 16 \text{ mm} \\ x_s = 5 \text{ mm} \\ D = 50 \text{ mm} \end{cases} \quad (12)$$

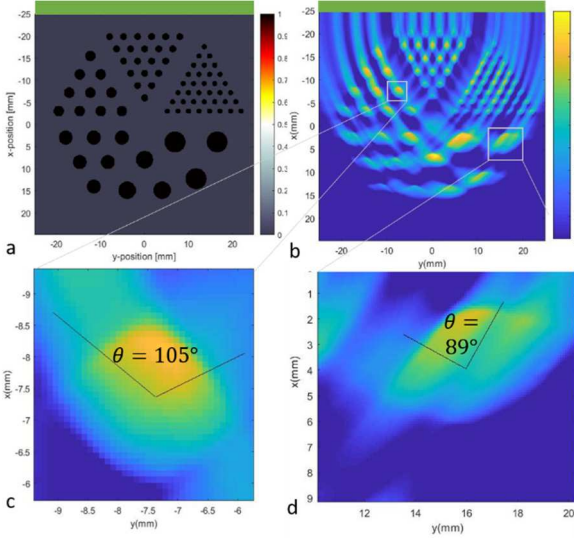


Fig. 5. Derenzo phantom and image reconstructed with planar transducers geometry. (a) Derenzo phantom and linear array transducers. (b) reconstructed image. (c,d) Magnified areas of the marked parts of the (b).

B. Circular Geometry

The dimensions of the imaging system and the phantoms in this section are exactly the same as the previous one, and the only difference between this part and the previous part is the arrangement of the transducers. for the limited-view mode, 120 transducers are located on a circular arc at a 240-degree angle with radius 25mm, shown in Fig. 6a with the Derenzo phantom. for full-view mode, 180 transducers are located on a circular arc at a 360-degree, shown in 6.b with the Derenzo phantom. Fig. 6c and 6d are reconstructed image of Fig. 6a and 6b, respectively. As can be seen, many of the points in Fig. 6a and 6b are not well reconstructed in Fig. 6c and 6d. It should be mentioned that, for better comparison these figures are normalized. From the figures, it can be seen that the more range of blind angles the phantom has, the lower the quality of the reconstruction is become. In (9) an equation for finding the blind angle of each source depending on its location is presented. To prove the equations of section II.B, we calculate the equations for the two parts of the Derenzo Phantom in Fig. 6c and 6d. Two parts of Fig. 6.c are selected, which are shown in Fig. 6e and 6h, respectively. These parts of Fig. 6d are shown in Fig. 6f and 6i, respectively.

The coordinates of the photoacoustic source center considered in Fig. 6e, 6f and $D_1(y_1, x_1)$ and $D_2(y_2, x_2)$ are

given in (13). By substituting (13) in (7), (8) and (9) we will obtain, $\theta_{\text{blind}} = 175$ degrees. If we subtract Fig 6e from 6f, we get Fig. 6g. In Fig. 6g as can be seen in some places the amplitude is more and in some places it is lower. In Fig. 6g, the low amplitude values indicate the regions fall in the acceptance angle and the high amplitude values indicate the regions fall in the blind angle and thus not reconstructed carefully in the limited-view structure. If we separate the parts of Fig. 6g that are more than 30% different from each other, we will see that the value of the points that differ by more than 30% is an arc from the circle at an angle of about 175 degrees, which corresponds to the result of the equations.

$$\begin{cases} y_s = -8.3 \text{ mm. } x_s = 8.5 \text{ mm} \\ y_1 = -25 \text{ mm. } x_1 = 0 \text{ mm} \\ y_2 = 12.4 \text{ mm. } x_2 = 21.4 \text{ mm} \end{cases} \quad (13)$$

The coordinates of the photoacoustic source center considered in Fig. 6h, 6i and $D_1(y_1, x_1)$ and $D_2(y_2, x_2)$ are given in (14). by substituting (14) in (7), (8) and (9) we will obtain, $\theta_{\text{blind}} = 144.5$ degrees. like before, if we subtract Fig 6h from 6i, we get Fig. 6j. If we separate the parts of Fig. 6j that are more than 30% different from each other and put them in Fig. 6f, we will see that the value of the points that differ by more than 30% is an arc from the circle at an angle of about 144.5 degrees, which again corresponds to the result of the equations.

$$\begin{cases} y_s = 4 \text{ mm. } x_s = 15 \text{ mm} \\ y_1 = -25 \text{ mm. } x_1 = 0 \text{ mm} \\ y_2 = 12.4 \text{ mm. } x_2 = 21.4 \text{ mm} \end{cases} \quad (14)$$

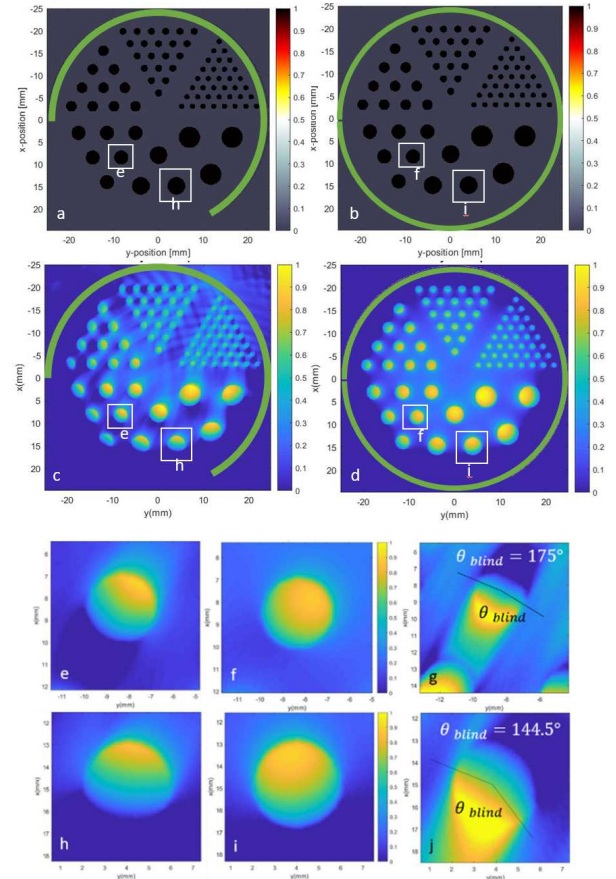


Fig. 6. Derenzo phantom and image reconstructed with limited-view and full-view circular transducers geometry. (a) Derenzo phantom and full-view circular transducers geometry. (b) Derenzo phantom and limited-view circular transducers geometry. (c, d) Image reconstructed of (a, b) respectively. (e, f) Magnified areas of the marked parts of the (c, d). (g) Absolute magnitude of differences (e) and (f). (h, i) Magnified areas of the marked parts of the (c, d). (j) Absolute magnitude of differences (h) and (i).

As shown in Fig. 6, depending on the position of the photoacoustic source and the view parameters in the circular arrangement, each photoacoustic source has a different blind angle, which is in accordance with the above equations. It should be noted that in a circular PACT reconstruction, the farther we go from the center, the smaller the amplitude of the reconstructed image would become. This effect although is normal, is different from the blind-angle discussion that discussed here. To study limited-view images, we should compare full view and limited-view images together, and comparing the limited-view image with the original phantom is a mistake, which is in accordance with the process performed in Fig. 6.

IV. CONCLUSION

In this paper, we introduced a concept called blind angle for photoacoustic computed tomography (PACT) that have planar or circular limited-view geometry. It is assumed that transducers used are point-like and the PACT system structure is two dimensional and the bandwidth is infinite. Equations in this paper are presented for different geometries that can be used to determine the blind angle for each source inside the chamber. The simulation results shown that the accuracy of these equations is very high. The importance of this work is that in many clinical and preclinical applications of photoacoustic tomography (PAT), we inevitably have to use linear or circular limited-view geometries because access to tissue is not possible from all directions. With using these equations, it is possible to specify the blind angles for each part in any photoacoustic imaging system with any desired geometry and by knowing the blind angles of each source, the quality of the reconstructed images can be improved. The results showed that in a linear geometry, as we move away from the horizon on which the transducers are arranged, the blind angle of each point increases and consequently the image quality decreases. For the limited-view layout, considering its angle, in particular, the results can be obtained, which are fully examined in this paper.

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